VNR Vignana Jyothi Institute of Engineering

Civil Engineering | Course Code: CE301

Structural Analysis & Indeterminate Systems - Laboratory Manual & Practi

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Syllabus Focus: CE301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Structural Indeterminacy & Energy Methods

" Syllabus Scope: Static and kinematic indeterminacy of beams, frames, and trusses, strain energy, Castig

Unit 2: Slope-Deflection Method for Beams & Frames

" Syllabus Scope: Slope-deflection equations, continuous beams with support settlements, non-sway and swa

CE301: Structural Analysis & Indeterminate Systems - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Moment Distribution Method Hardy Cross

" Stiffness factors, carry-over factors, distribution factors, continuous beams with sinking supports, single-

Unit 4: Influence Lines for Moving Loads

" Influence line diagrams ILD for reactions, shear force, and bending moments in determinate beams, Muller-Bre

Unit 5: Matrix Methods of Structural Analysis

" Flexibility matrix method force method, Stiffness matrix method displacement method, element coordinate tran

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Verification of Castiglianos Second Theorem for Deflection of a Cantilever Truss

Lab 2: Experimental Determination of Influence Lines for a Continuous Beam Model

Lab 3: Structural Analysis and Deflection Verification of Redundant Portal Frames

Lab 4: Computer-Aided Structural Modeling and Stress Analysis of High-Rise Frames using STAAD.Pro

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* C.S. Reddy - Basic Structural Analysis McGraw-Hill 3rd Ed

* R.C. Hibbeler - Structural Analysis Pearson 9th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students